

KELECHI IGWE

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[GitHub](#) || [Google Scholar](#) || [ResearchGate](#) || [LinkedIn](#)

EDUCATION

Kansas State University, Manhattan KS

- Doctor of Philosophy (Ph.D.) in Biological and Agricultural Engineering Expected 2027
- Master of Science (M.Sc.) in Biological and Agricultural Engineering (GPA: 4.0/4.0) 2023

Thesis title: Modeling Agricultural Crop Water Demands and Sustainable Water Management Under Extreme Climate Conditions in Western Kansas

Advisor: Dr. Vaishali Sharda

Federal University of Technology, Owerri, Nigeria

- Bachelor of Engineering (B.Eng.) in Agricultural Engineering (Soil & Water Eng.) 2018

TECHNICAL SKILLS

Software: Proficient in Python and R Programming, High Performance Computing, ArcGIS Pro, Decision Support System for Agrotechnology Transfer (DSSAT), Google Earth Engine (GEE) code editor, SWAT, AGISOFT, and ENVI software.

Analytical: Remote Sensing, Machine learning, Spatial-Temporal modeling and analysis, Crop and Hydrological modeling

RESEARCH EXPERIENCE

Graduate Research Assistant

2022 – Present

Advisor: Dr. Vaishali Sharda

Department of Biological and Agricultural Engineering, Kansas State University

- Developing a multiscale remote sensing and modeling framework for climate-smart agricultural water management using UAV multispectral and thermal imagery, and machine learning models (funded by National Science Foundation)
- Designing data-driven early-warning systems to detect Maize water stress and optimize irrigation management across field, county, and regional scales (funded by National Science Foundation)
- Investigated crop adaptation strategies under extreme climate conditions in arid and semi-arid regions (Funded by Carl R. Ice College of Engineering, Kansas State University)
- Implemented adaptive irrigation strategies for improving crop productivity under extreme climate conditions, using the Decision Support System for Agrotechnology Transfer (DSSAT) Maize model. (Funded by Carl R. Ice College of Engineering, Kansas State University)
- Applied machine-learning combined with satellite remote-sensing technology to estimate future crop water needs and climate-associated impacts in western Kansas (Funded by Carl R. Ice College of Engineering, Kansas State University)
- Developed automated data-processing workflows in R and Python to extract, analyze, and visualize large simulation outputs from crop and environmental models.
- Conducted field data collection and sensor deployment to monitor soil moisture dynamics and environmental conditions supporting model calibration and validation (Funded by Kansas Soybean Commission)

NASA DEVELOP Intern

Jan - April 2025

Contractor to National Aeronautics and Space Administration (NASA) through Analytical Mechanics Associates (Remote)

- Applied statistical modeling with earth observation data to assess environmental drivers of cotton quality
- Developed and implemented Python scripts to automate the extraction and processing of multi-satellite data products using application programming interfaces (APIs)
- Performed spatial analysis and visualization of geospatial data using geographic information systems software (ArcGIS Pro)

- Collaborated with multi-disciplinary team to create scientific posters and presentations to effectively communicate research findings.

ADVANCED EDUCATION AND PROFESSIONAL TRAINING

- 2026 Bayesian Modeling of Designed Experiments. Institute for Digital Agriculture and Advanced Analytics. Manhattan, KS. March 2-5, 2026
- sUAS Remote Pilot Training (FAA Part 107). Federal Aviation Administration. April 4, 2025.
- 2024 I-GUIDE Summer School on "Leveraging AI for Environmental Sustainability". Institute for Geospatial Understanding through an Integrative Discovery Environment. University Consortium for Atmospheric Research, Boulder, CO. August 5-9, 2024.
- Oxford School of Climate Change course, Hilary Term 2024. The Oxford Climate Society. Jan.18 – Mar. 7, 2024.
- Irrigation Association 2023 University Classes for irrigation professionals. Irrigation Show and Education Week. San Antonio, Texas. Nov. 27 - Dec. 1, 2023
- Decision Support System for Agrotechnology Transfer (DSSAT) 2023 International Training Program. The University of Georgia – Griffin Campus. May 15-21, 2023.
- Graduate Certificate in Geographic Information Science. Department of Geography, Kansas State University, Manhattan, KS. January 2022 - May 5, 2023

RELEVANT COURSES

Principles of Irrigation: Agriculture, Irrigated soils: Essentials of the soil-water relationship; Geographic Information Systems I & II, Remote Sensing and Environment; Drone Imagery Collection, Mapping, and Analysis; Introduction to Statistical Methods for the Sciences, Regression and Analysis of Variance; Bayesian Modeling of Designed Experiments; Assessing Crop Production, Water and Nutrient Management, Climatic Risk and Environmental Sustainability with Simulation Models; Climate Change and the Biosphere, The Science and the IPCC, Climate Action from above and below.

SYNERGISTIC ACTIVITIES

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| • Certified Journal Reviewer, Elsevier's Journal of Agricultural Water Management | 2025 – Present |
| • Certified Journal Reviewer, American Society of Agricultural and Biological Engineers | 2023 – Present |
| • Globe IVSS Judge, 2025 GLOBE International Virtual Science Symposium (IVSS) | 2025 |
| • Student Essay Reviewer, Agriculture Future of America | 2025 – 2026 |
| • Reviewer, NASA's Minority University Research and Education Project (MUREP) | 2025 |

LEADERSHIP AND COMMUNITY SERVICE

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| • Illinois Science and Technology Coalition (ISTC Education), <i>Mentor</i> | Sept. 2025 - Present |
| • Africans in Environmental Science Mentorship Program, <i>Mentor</i> | Aug. 2025 - Present |
| • African Network Group of the American Society of Agricultural and Biological Engineers, <i>Executive Student Representative</i> | 2023 - Present |
| • Young Professionals Community of the ASABE, <i>Member</i> | 2023 - Present |
| • HandsOn K-State, 2024 MLK Day of Service, <i>Student volunteer</i> | Jan. 2024 |
| • Anthony Middle School, Manhattan KS, <i>Grader for DEN161 Expo</i> | Dec. 2023 |
| • Staley Leadership School, Kansas State University, KS, <i>Volunteer for the yearly Mandela Washington Fellowship</i> | 2022 - 2023 |
| • African Students Union, Kansas State University, <i>ASUed Initiative Volunteer</i> | 2022 - 2023 |
| • Helping International Students, Manhattan KS, <i>Student Leader</i> | 2022 - 2023 |
| • Prime Africans Network, <i>Program Team Lead</i> | 2018 - 2021 |
| • Annual Nigeria Innovation Summit, <i>Team Member</i> | Oct. 2019 |
| • Nigerian Christian Corpsers' Fellowship, Ido Zone, Oyo State Nigeria, <i>Coordinator</i> | 2019 - 2020 |
| • Christ Ambassadors Students' Outreach, FUTU, Owerri Nigeria, <i>President</i> | 2016 - 2017 |

PROFESSIONAL MEMBERSHIPS/AFFILIATIONS

- American Society of Agricultural and Biological Engineers, *Member (M1062832)*

- International Society of Precision Agriculture, *Member (ISPA-02601)*
- Environmental and Water Resources Institute (EWRI)
- American Society of Civil Engineers (ASCE), *Member (Reference Number: 1047356410)*
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- Institute of Electrical and Electronics Engineers (IEEE), *Member (102061132)*
- IEEE Geoscience and Remote Sensing Society, *Member (102061132)*
- The Irrigation Association, *E3 Learner*
- Nigerian Society of Engineers, *Student Member (S40068/18)*

HONORS/AWARDS

- Finalist, K-State Three Minute Thesis (3MT) Competition, Kansas State University **2026**
- 2026 NIWR Student Travel Award Winner, National Institutes for Water Resources **2026**
- Governor's Outstanding Graduate Research Award (Graduate Runner-up), 2025 Student Poster Contest-Governor's Water Conference, Kansas Water Office. **2025**
- Winner, Capitol Graduate Research Summit, Research and the State, Kansas State University Graduate School. **2025**
- First Place Student Research Award 2024, African Network Group of the American Society of Agricultural and Biological Engineers **2024**
- First Place Student Research Award 2024, Association of Agricultural, Biological and Food Engineers of Indian Origin **2024**
- Selected 2024 Scholar, NSF Institute for Geospatial Understanding through an Integrative Discovery Environment (I-GUIDE) **2024**
- One of selected Hilary Term 2024 Cohort, Oxford School of Climate Change from over 5000 competitive applications **2024**
- One of 20 winners, Irrigation Exposure, Experience & Education (E3) Program Class of 2023 **2023**
- Workshop Travel Award, North Central Region Water Network, Denver Colorado **2023**
- Student Water Conference Travel Award, South Dakota State University **2023**
- Third place Student Research Award (Masters category), Association of Agricultural, Biological and Food Engineers of Indian Origin **2023**
- Graduate Travel Award, Graduate Student Council, Kansas State University **2023**

PUBLICATIONS

In Review/Preparation

- **Igwe, K., & Sharda, V. (2026).** High-Resolution Estimation of Irrigation Water Use Using Satellite-Based Soil Moisture Data: County-Level Assessment Across Western U.S. Crop Production Systems (Manuscript in preparation)
- **Igwe, K., Goodin, D.G., Eric, J.A. Ramos, G., & Sharda, V. (2026).** High-Resolution Mapping of Early Water Stress in Irrigated and Dryland Maize Using Radiative Transfer Modeling and Multi-Modal Remote Sensing (Manuscript in preparation)
- **Igwe, K., Singh, M., Kaur, M., Monavarian, A., Ramos, G., Madin, M., Onyekwelu, I., Rawat, M., & Sharda, V. (2026).** Advancing crop adaptation to climate change in arid and semi-arid regions: a systematic review of past, present and potential future approaches (In Review)

Published

- Dar, M. U. D., Onyekwelu, I., **Igwe, K., Singh, M., & Sharda, V. (2025).** Future Climate Projections for Nebraska: Insights from CMIP6 on Precipitation and Temperature Trends. *Earth Systems and Environment*. <https://doi.org/10.1007/s41748-025-00977-5>
- **Igwe, K., Onyekwelu, I., & Sharda, V. (2025).** Adaptation strategies for deficit irrigation management under extreme climate conditions. *Agricultural Water Management*, 313, 109506. <https://doi.org/10.1016/j.agwat.2025.109506>
- Andrade, L., Grener, N., **Igwe, K., & Osei, H. (2025).** Texas & Georgia Agriculture : Assessing the Drivers of Cotton Quality and Yield for Improved Crop Forecasting in the Southern United States . 2025 DEVELOP Closeout.

https://ntrs.nasa.gov/api/citations/20250003123/downloads/2025Spring_GSFC_TXandGAAG_Pres_FD-Final.pdf

- Dar, M.U., Onyekwelu, I., Guanes, A., **Igwe, K.**, Singh, M., & Sharda, V. (2024). Projected Changes in precipitation and air temperature across Nebraska using CMIP6 Global Climate Models. (Under Review)
- **Igwe, K. E.** (2023). Modeling agricultural crop water demands and sustainable water management under extreme climate conditions in Western Kansas (Masters thesis). URI: <https://hdl.handle.net/2097/43539>
- **Igwe, K.**, Sharda, V., & Hefley, T. (2023). Evaluating the Impact of Future Seasonal Climate Extremes on Crop Evapotranspiration of Maize in Western Kansas Using a Machine Learning Approach. *Land*, 12(8), Article 8. <https://doi.org/10.3390/land12081500>
- Wiederstein, T., Sharda, V., Aguilar, J., Hefley, T., Ciampitti, I. A., Sharda, A., & **Igwe, K.** (2022). Evaluating spatial and temporal variations in sub-field level crop water demands. *Frontiers in Agronomy*, 0, 98. <https://doi.org/10.3389/FAGRO.2022.983244>

CONFERENCE PRESENTATIONS

Oral presentation

- **Igwe, K.**, Goodin, D.G., Eric, J.A. & Ramos, G. (2025). Leveraging AI and Remote Sensing to Enhance Water Stress Detection in Dryland and Irrigated Maize. 2025 UCOWR/NIWR Annual Water Resources Conference. University of Minnesota, Minneapolis, MN. Coming up by June 3, 2025 – June 5, 2025.
- Andrade, L., Grener, N., Igwe, K., & Osei, H. (2025). Texas & Georgia Agriculture: Assessing the Drivers of Cotton Quality and Yield for Improved Crop Forecasting in the Southern United States. 2025 NASA DEVELOP Closeout. April 10, 2025. GSFC, Maryland (Virtual)
- **Igwe, K.**, Onyekwelu, I., & Sharda, V. (2024). Adapting Deficit Irrigation to Extreme Climate Conditions. 2024 Annual International Meeting of the American Society of Agricultural and Biological Engineers, Anaheim, California. July 28-31, 2024
- **Igwe, K.**, Onyekwelu, I., & Sharda, V. (2024). Precision irrigation strategies for climate-resilient crop production and water resource management. 16th International Conference on Precision Agriculture. Manhattan, KS. July 21-24, 2024
- **Igwe, K.**, Onyekwelu, I., Sharda, V., Hefley, T., & Sheshukov, A. (2023). Investigating sustainable water management strategies for improving crop adaptation to extreme climate conditions in western Kansas. South Dakota Student Water Conference. Brookings, SD. October 10-11, 2023
- **Igwe, K.**, Sharda, V., & Hefley, T. (2023). Estimating crop water requirements under extreme climate conditions: A machine-learning approach. 2023 Annual International Meeting of the American Society of Agricultural and Biological Engineers, Omaha, Nebraska. July 9-11, 2023
- **Igwe, K.**, Sharda, V., & Hefley, T. (2023). Effects of climate extremes events on crop water demands. K-State Graduate Research, Arts, and Discovery (GRAD) Forum. March 9, 2023

Poster presentation

- **Igwe, K.**, & Sharda, V. (2025). Closing the Water Budget Gap: Leveraging Satellite-Based Insights for Accurate Irrigation Water Use Estimation across the Western U.S. Governor's Water Conference on the future of water in Kansas, Manhattan, KS. November 12-14, 2025.
- **Igwe, K.**, & Sharda, V. (2025). Closing the Water Budget Gap: Leveraging Satellite-Based Insights for Accurate Irrigation Water Use Estimation across the Western U.S. Research and the State, Manhattan, KS. October 28, 2025.
- **Igwe, K.**, & Sharda, V. (2025). A Hybrid Remote Sensing and Physics-Based Modeling Framework for Irrigation Water Use Estimation in Western US 2025 American Society for Photogrammetry and Remote Sensing (ASPRS) International Technical Symposium. Virtual. October 27-30, 2025.
- Igwe, K., Onyekwelu, I., & Sharda, V. (2025). Adaptation strategies for deficit irrigation management under extreme climate conditions. 15th European Conference on Precision Agriculture (ECPA 2025). Barcelona, Spain. June 29-July 03, 2025.

- **Igwe, K.,** Onyekwelu, I., & Sharda, V. (2024). Optimizing water management for sustainable agriculture in the face of extreme climate conditions. 2nd International Conference on Water Resources Management & Sustainability: Solutions for Arid Regions (WARMS 2024). Al Jaddaf, Dubai. February 26-28, 2024.
- **Igwe, K.,** Onyekwelu, I., & Sharda, V. (2023). Growing more with less: Smart irrigation for a sustainable future in Western Kansas. Governor's Water Conference, Manhattan, KS. November 14-16, 2023.
- **Igwe, K.,** Onyekwelu, I., & Sharda, V. (2023). Growing more with less: Smart irrigation for a sustainable future in Western Kansas. Research and State. Kansas State University, Manhattan, KS. October 24, 2023
- **Igwe, K.,** Onyekwelu, I., Sharda, V., Hefley, T., & Sheshukov, A. (2023). Enhancing crop resilience to extreme climate conditions through sustainable water management practices. DA3: Digital Ag & Advanced Analytics. Manhattan, KS. October 23, 2023
- **Igwe, K.,** Sharda, V., & Hefley, T. (2023). Use of remote sensing and machine learning techniques to study the impact of climate extremes of crop evapotranspiration. 14th European Conference on Precision Agriculture: Unleashing the potential of precision agriculture. P83. Bologna, Italy. June 29, 2023
- **Igwe, K.,** Sharda, V., Sheshukov, A., & Hefley, T. (2022). Modeling the impact of climate extremes on crop evapotranspiration. Governor's Water Conference on the Future of Water in Kansas, Manhattan, KS.

Workshops attended

- Satellite-Enabled Earth Observation & Digital Solutions for Agriculture (SEEDS). Seminar Series. University of Maryland. October 14-17, 2025 (Virtual)
- Climate, Water, Equity and Opportunity: A Workshop for all Land-grant Institutions hosted by the North Central Region Water Network. Denver, Colorado. October 2023
- Rural Resource Resiliency NSF Research Graduate Traineeship (R3NRT). SW Kansas Research Experience. July 25, 2022

RESEARCH GRANTS

Funded Awards

- 2025: Enhancing Early Detection of Crop Water Stress in Dryland and Irrigated Maize Using Single and Multi-Modal Water Stress Indices
Funder: Kansas State University (Graduate Research, Scholarly and Creative Activities, and Discovery – RSCAD)
Total Award: \$1,000 (Funded)
Role: Student Investigator
Collaborator: Vaishali Sharda

External Grants (Not Funded)

- 2025–2027: Integrating Remote Sensing and Machine Learning for Improved Quantification of Annual Irrigation Water Use Across the Western United States
Program: Future Investigators in NASA Earth and Space Science and Technology (FINESST)
Funder: NASA Science Mission Directorate
Solicitation: Research Opportunities in Space and Earth Sciences (ROSES-2024)
Total Amount Requested: \$150,000 (Not Funded)
Role: Future Investigator (FI)
Co-Investigator: Vaishali Sharda
- 2026–2027: Coupled Remote Sensing and Land Surface Data Assimilation for Agricultural Water Management in Semi-Arid Western United States
Funder: Corteva Halo Science
Total Award: \$100,000 (Not Funded)
Role: Graduate Student Researcher
PI: Vaishali Sharda
- 2026–2027: RAPS – Regenerative Agriculture Pathways for Sustainability
Program: Building Resilience in Agriculture

Funders: National Geographic Society and PepsiCo
Total Award: \$150,000 (Not Funded)
Role: Graduate Student Researcher
PI: Vaishali Sharda